

What is M&I

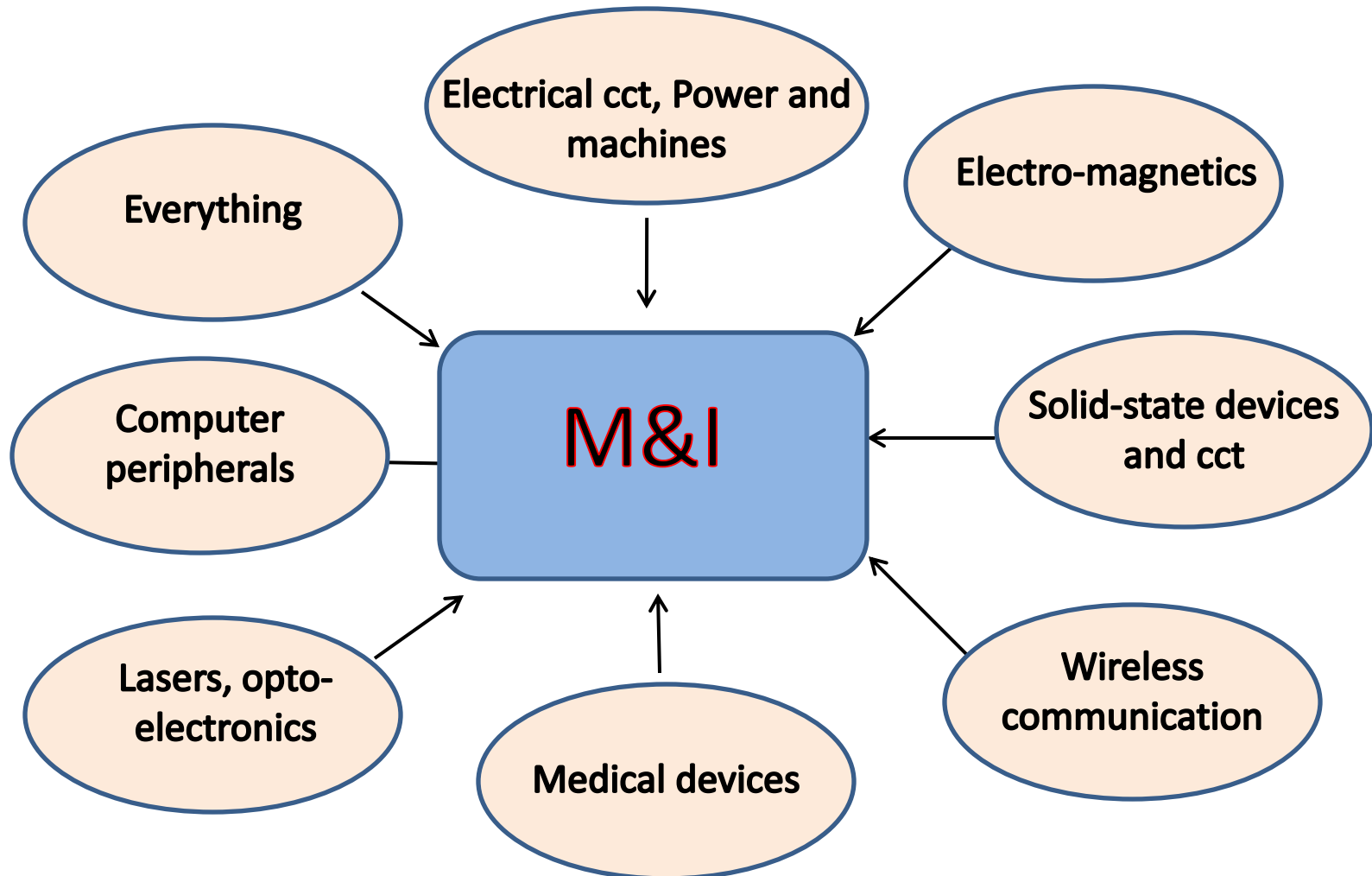
Measurement (M): Result of comparison between a quantity (of unknown magnitude) and a predefined standard

Instrument (I): A device for determining the magnitude of a quantity or variable

- M&I is very important for each branches of science and engineering
- EEE 427 covers obviously only the M&I related to EE Engineering

What is NOT included in M&I?

Ans: **NOTHING**



WELCOME TO THE ENDEAVOR

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Applications of Measurement Systems

Applications are generally classified into three major classes:

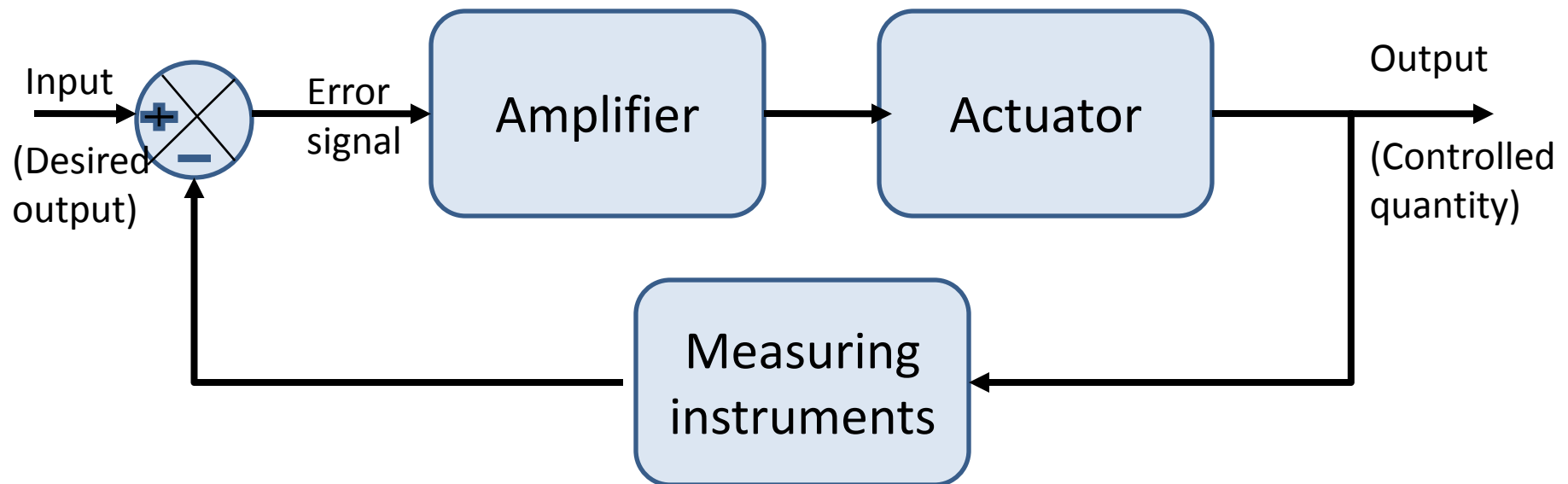
1. Monitoring of processes and operations
2. Control of processes and operations
3. Experimental engineering analysis

1. Monitoring of processes and operations

- measuring instrument is used to keep track of some quantity.
- e.g. thermometer, barometer, radar used by weather department
- e.g. water, gas, electric meters in home

2. Control of processes and operations

- The most important class of measurement application (automatic control systems)
- Feedback control system:



3. Experimental engineering analysis

- Part of engineering design, development and research
- two basic ways of solving engineering problems: Theory and Experimentation
- Most problems require blend of theory and experiment
- For example,
 - Experimental validation of theoretical prediction/simulation
 - Formulation of empirical equations

Functional Elements of a Measurement System

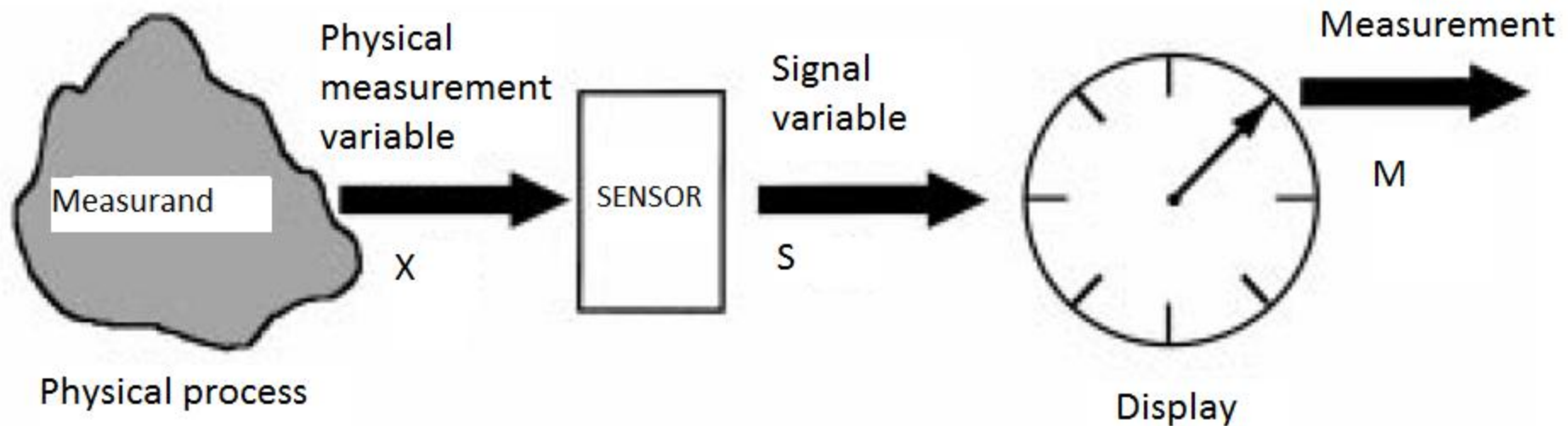


Figure: Simple instrument model

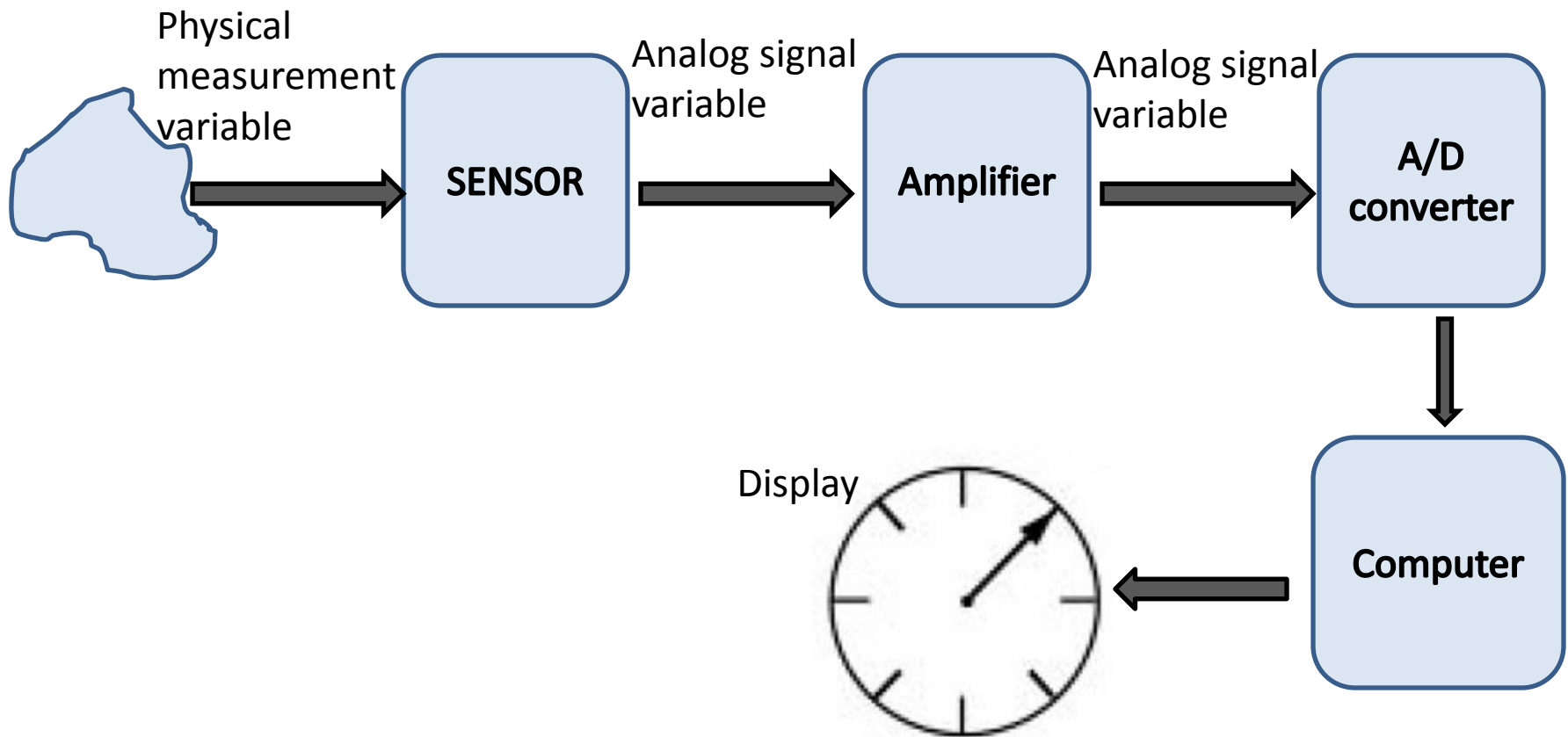


Figure: An instrument with amplifier, A/D converter and computer interfacing

Generally, measurement systems contain three functional elements:

1. Primary sensing element
2. Signal conditioning element and
3. Data presentation element

Each element is made up of a distinct component or **a group of components** which perform the required function

1. Primary sensing element

Variously known as detector/sensor or transducer

2. Signal conditioning element

- sometimes, the raw signal output from the sensor is not suited to the desired performance of the system
- For example: filtering noise, amplifying signal and A/D conversion
- Many instruments do not need any signal conditioning element

3. Data presentation element

- The quantity under measurement has to be conveyed to a personnel or to the system for monitoring, control, or analyses purposes.
- The data conveyed must be intelligible to the personnel/system
 - manual monitoring: visual display (LED/LCD, pointer etc)
 - recording: magnetic tape, printers, computer etc
 - control and analysis: computer, microcontroller

IMPORTANT TERMINOLOGIES

Passive and Active Sensors:

Passive Sensors

Passive sensors do not add energy as part of the measurement process but may remove energy in their operation

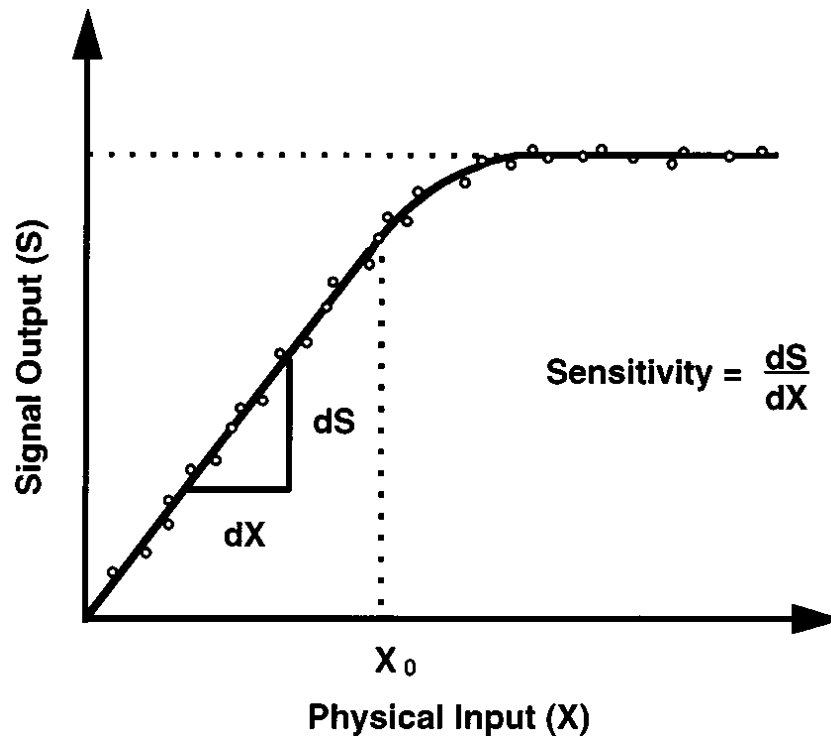
Active Sensors

Add energy to the measurement environment as part of the measurement process.

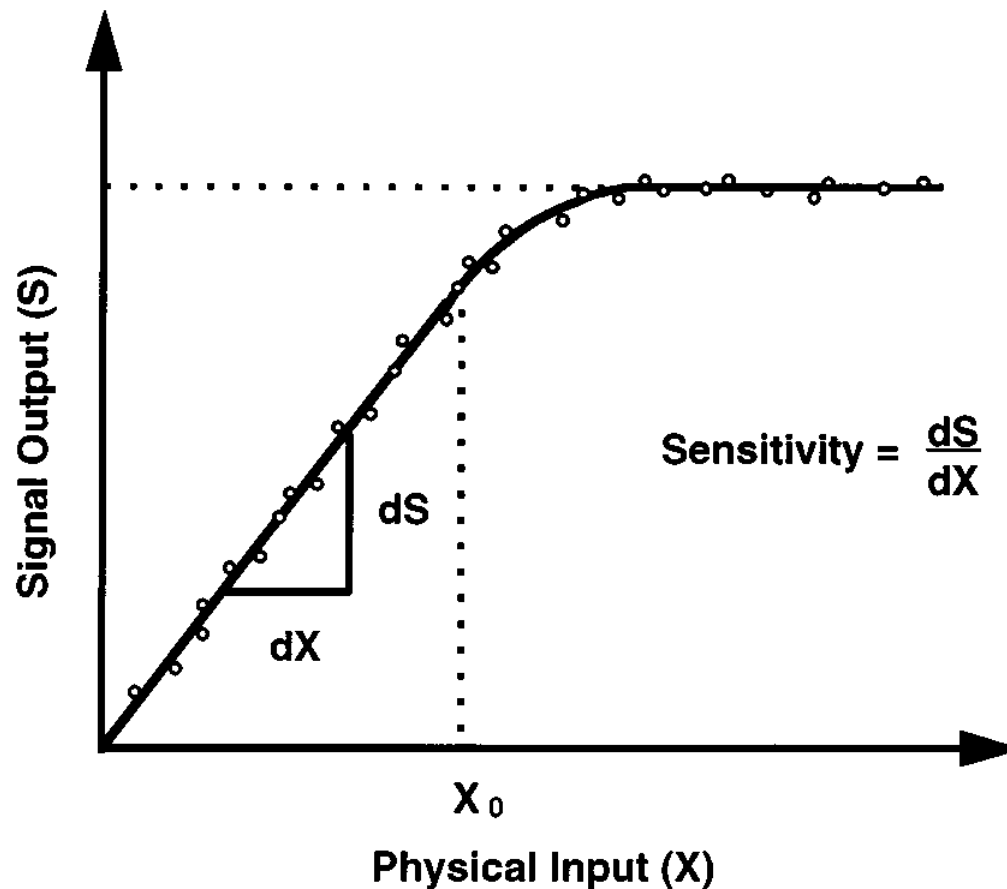
Example: A radar or sonar system, where the distance to some object is measured by actively sending out a radio (radar) or acoustic (sonar) wave to reflect off of some object and measure its range from the sensor.

Calibration: The relationship between the physical measurement variable (input) and the signal variable (output) for a specific sensor is known as the calibration of the sensor. Typically, a sensor (or an entire instrument system) is calibrated by providing a known physical input to the system and recording the output.

Calibration Curve:



Sensitivity: Ratio of output signal or response of the instrument to a change of input or measured variable



Types of errors:

Gross errors: largely human errors, e.g. misreading, incorrect adjustment and improper application of instruments

Systematic errors (bias): shortcomings of the instruments, e.g. effects of environment (temperature, humidity etc) on the instrument, change in instrument response due to aging.

- Use of compensation can reduce/remove some systematic error
- periodic recalibration can prevent systematic error

Random error (noise): caused by the noise introduced during the measurement process and random variation of input itself

- taking a large number of readings and averaging them can reduce the effect of noise

Accuracy: Closeness with which an instrument reading approaches the true value of the variable

- It is affected by the systematic error (bias) of the instrument

= true value – statistical mean of the measurements

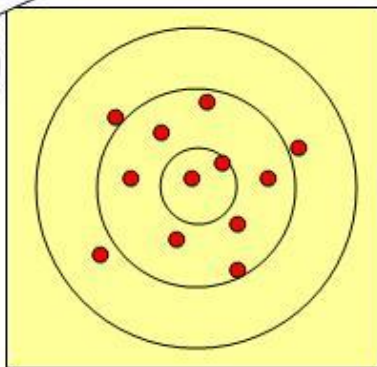
Precision: A measure of the reproducibility (or spread) of the measurements. It is a measure of the degree to which successive measurements differ from one another.

- It is affected by the random error (noise) of the instrument

= standard deviation (σ) of the measurement

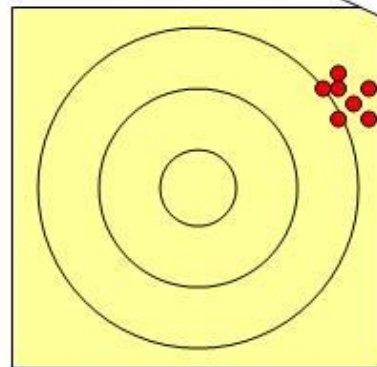
Accuracy vs Precision

*Measure of
bias*

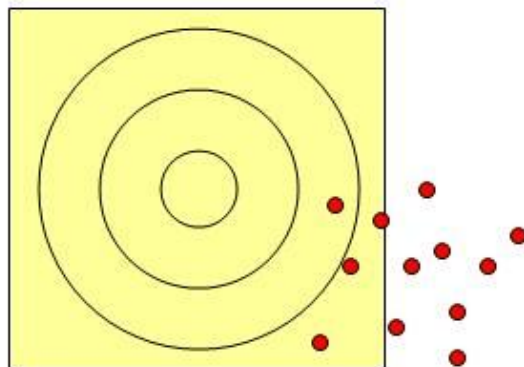


Accurate but not Precise

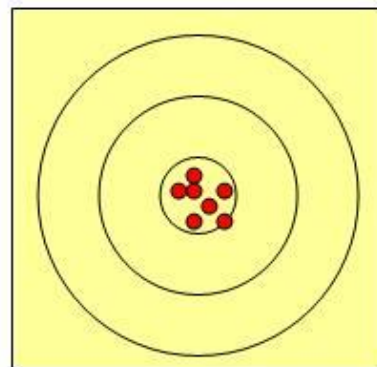
*Measure
of spread*



Precise but not Accurate



Not Accurate or Precise



Both Accurate & Precise

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Classifications of Instruments

Basic classification;

1. **Mechanical instruments-** They are very reliable for static and stable conditions.

Disadvantage: unable to respond rapidly to measurement of dynamic and transient conditions.

2. **Electrical instruments-** This method is more rapid than mechanical methods. They normally depends upon a electro-mechanical meter movement

3. **Electronic instruments:-** Have very fast response.

e.g., cathode ray oscilloscope (CRO) is capable to follow dynamic and transient changes of the order of few nanoseconds (10^{-9} s).

Classifications of Electrical Instruments

This classification is based on the various effects of electricity:

- **Magnetic effect:** Used in deflection-type ammeters, voltmeters, wattmeters etc.
- **Heating effect:** Used in ammeters and voltmeters
- **Chemical effect:** Used in dc ampere hour meters
- **Electrostatic effect:** Used in voltmeters
- **Electromagnetic induction effect:** Used in ac ammeters, voltmeters, wattmeters

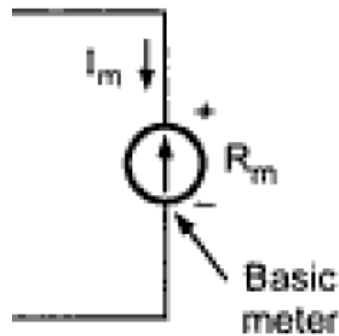
Magnetic and the electromagnetic induction effects are utilized in most cases

Classification based on the Nature of their Operations

- **Indicating instruments:** Indicated by means of a mechanical pointer which moves on a scale, or digital display. e.g., ammeter, voltmeter, wattmeter
- **Recording instruments:** Record continuously the variation of any electrical quantity with respect to time.
- **Integrating instruments:** Record the consumption of the total quantity of electricity, energy etc., during a particular period of time. Ampere-hour meter: kilowatthour (kWh) meter

Circuit Symbol of Basic PMMC Meter

Symbols of basic PMMC meter may **NOT** be unique

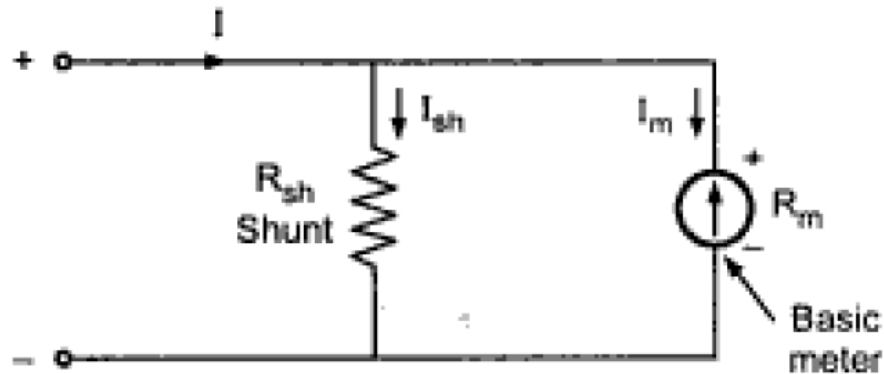


$I_m \rightarrow$ Full-scale deflection current through the basic meter

$R_m \rightarrow$ resistance of the basic meter

DC PMMC Ammeter

I_m can be only about 1 mA. For large current measurements, a shunt resistance R_{sh} must be connected to the basic meter.



$$I_{sh} R_{sh} = I_m R_m$$

$$\therefore R_{sh} = \frac{I_m R_m}{I_{sh}}$$

$$\text{but } I_{sh} = I - I_m$$

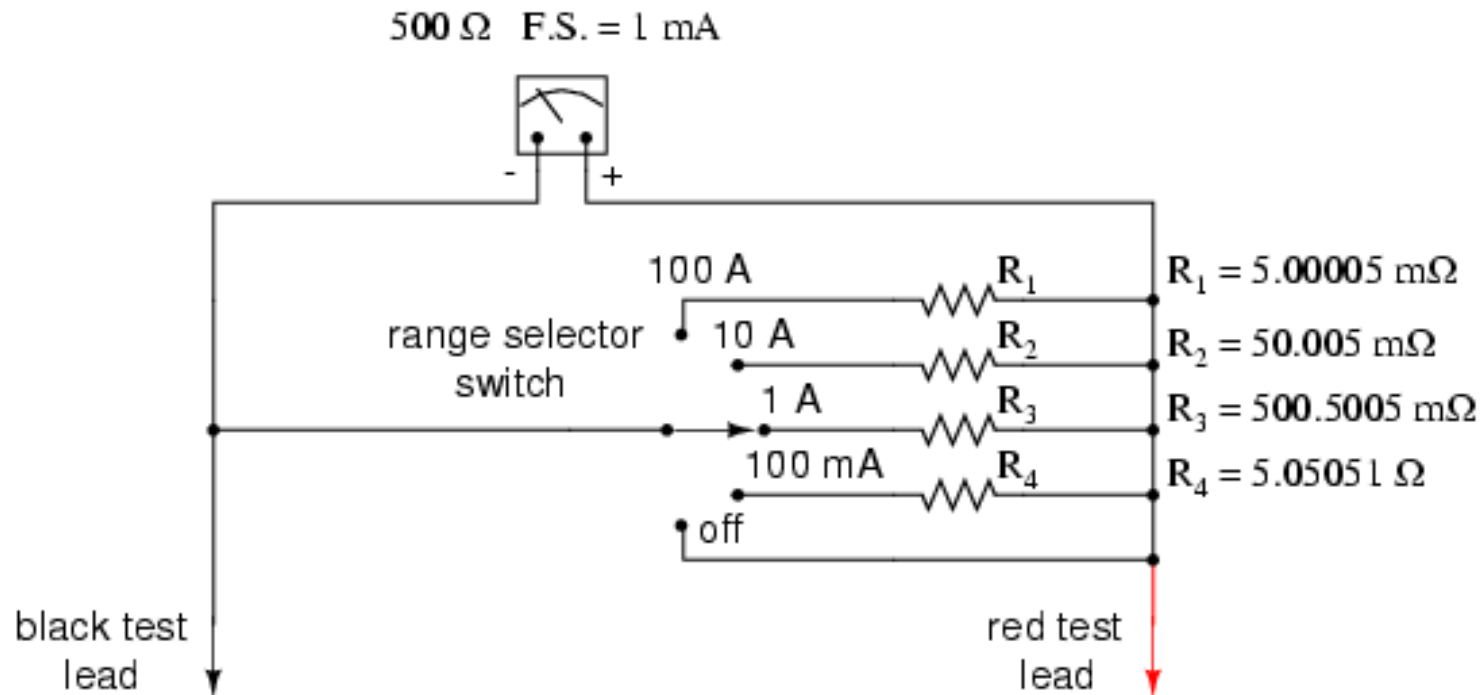
$$\therefore R_{sh} = \frac{I_m R_m}{(I - I_m)}$$

$$\therefore \boxed{R_{sh} = \frac{R_m}{m - 1}} \quad \text{where } m = \frac{I}{I_m}$$

Multirange Ammeter

Various values of resistances can be used by means of suitable switching arrangement, to increase the number of current ranges.

1. Simple arrangement:

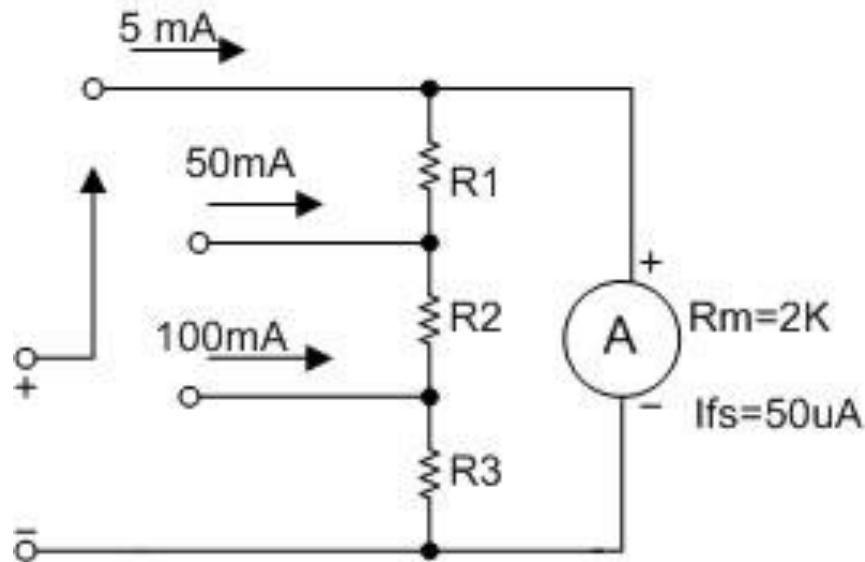


Disadvantages:

1. During the changeover of switch, the shunt is momentarily removed from the meter. Then, the full current flows through the meter coil. This can damage the coil.
2. The contact resistance of the switch is in series with the shunt. It can be of variable nature. So, the ammeter indication may not be accurate.

Multirange Ammeter

2. **Ayrton Shunt/universal shunt Arrangement:**



- Eliminates the possibility of momentary disconnection of the shunt.
- The contact resistance of the switch is external to the the shunt and meter.

Ammeter Loading Effect

If the internal resistance of the meter is a large portion of the load resistance of the circuit under measurement, an effect known as meter-loading occurs.

- Meter loading occurs when the insertion of the meter into the circuit changes the voltage/current values of that circuit. We say that the meter is loading the circuit.

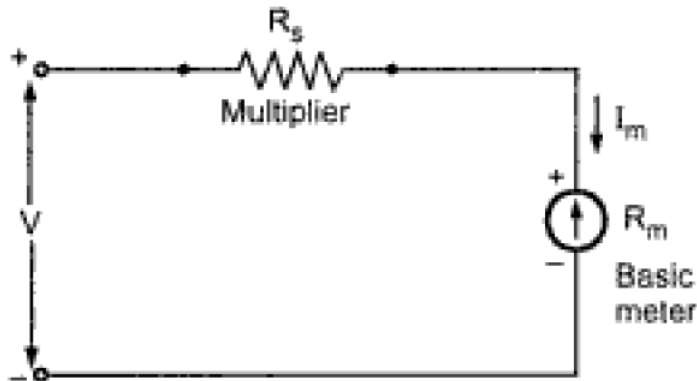
EX: Shown in the whiteboard

DC PMMC Voltmeter

Let, $I_m = 1 \text{ mA}$, and $R_m = 1000 \Omega$

So the basic meter can be connected across a maximum of 1 V only.

-To increase the range, resistors (R_s) are connected in series with the basic meter. Sometimes these resistors are called 'multipliers'.



Applying KVL,

$$V = I_m (R_m + R_s)$$

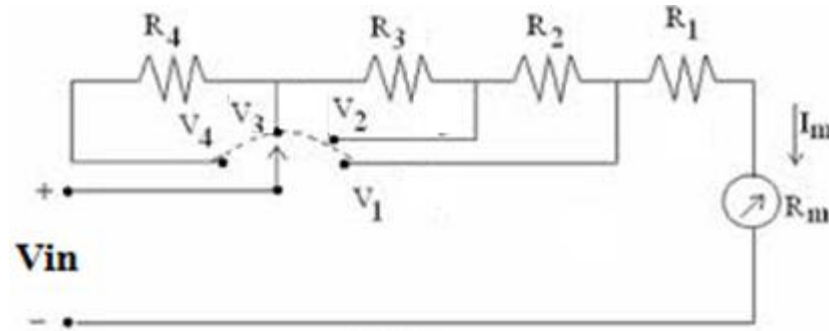
$\therefore$

$$R_s = \frac{V}{I_m} - R_m$$

$$m = \text{Multiplying factor} = \frac{V}{v}$$

$$m = 1 + \frac{R_s}{R_m}$$

Multirange Voltmeter



EX:

A 1 mA meter movement with an internal resistance of $50\ \Omega$ is to be used in a 0-1 V, 0-10 V, 0-50 V, and 0-100 V ranges in the arrangement shown above. Find the value of the required multiplier resistances.

Ans: $R_1 = 950\ \Omega$
 $R_2 = 9\ \text{k}\Omega$
 $R_3 = 40\ \text{k}\Omega$
 $R_4 = 50\ \text{k}\Omega$

Voltmeter Sensitivity

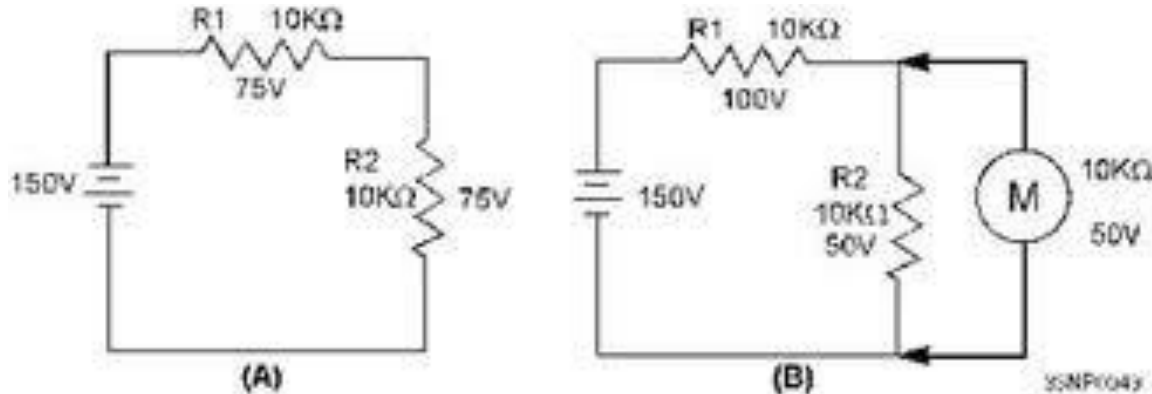
- Sensitivity of a voltmeter is given in Ω/V or A^{-1}
- Obtained by dividing the sum of R_m and R_s by the full-scale reading in volts.

$$\text{Sensitivity, } S = \frac{R_m + R_s}{V_{in}}$$

- In terms of full-scale deflection current,

$$\text{Sensitivity, } S = \frac{\text{Ohm}}{\text{Volt}} = \frac{1}{\text{Amp}} = \frac{1}{I_m}$$

Voltmeter Loading Effect



- It is caused principally by low- S instruments.
- So, S is an important factor when selecting a meter for a particular measurement
- For voltage measurement in high-resistance circuits, high S voltmeter should be used to prevent the loading effect

Voltmeter Loading Effect

1.8 In the circuit shown in Figure Q1.8, the voltage across the resistor $R_L (= 500 \text{ k}\Omega)$ is to be measured first by using a voltmeter with sensitivity of $100 \text{ k}\Omega/\text{V}$ and then with a voltmeter of sensitivity of $500 \text{ k}\Omega/\text{V}$. Calculate the reading of the voltmeter in each case and the % error in the measurement.

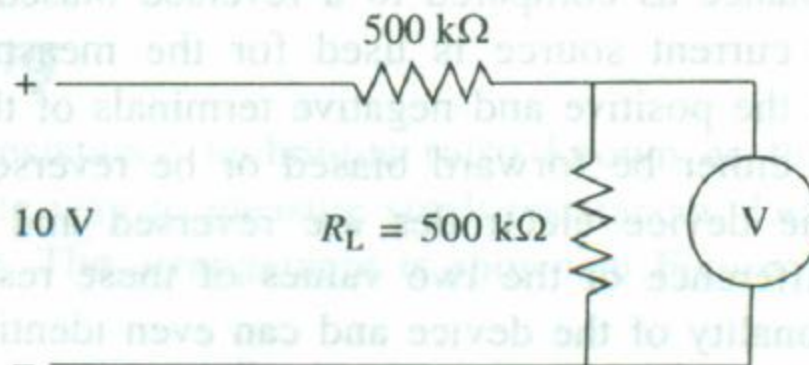
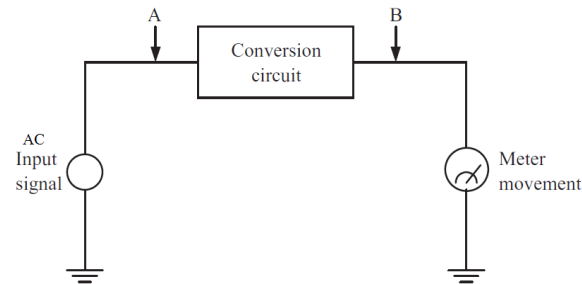


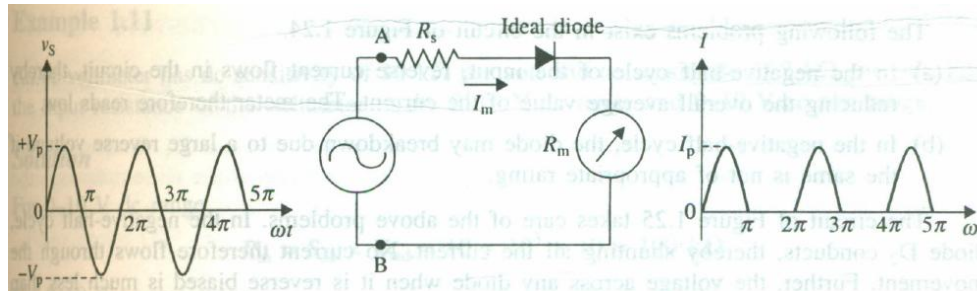
Figure Q1.8

Solve the Problem

Rectifier-based AC Meters



Half-wave rectifier AC Meter:



$$I_p = \frac{V_p}{R_s + R_m}$$

$$I_{av} = \frac{I_p}{\pi} = 0.318 I_p$$

To calculate R_s . The multiplier resistance, R_s , can be expressed in terms of the rms voltage, V_{rms} , average current and meter resistance as follows:

$$\begin{aligned} V_{rms} &= 0.707 V_p = 0.707 I_p (R_s + R_m) = \frac{0.707}{0.318} I_{av} (R_s + R_m) \\ &= 2.22 I_{av} (R_s + R_m) \end{aligned}$$

Therefore,

$$R_s = 0.45 \frac{V_{rms}}{I_{av}} - R_m$$

Note that we have considered an ideal diode. In a practical diode, the diode cut-in voltage and diode resistance should also be considered.

Example 1.10

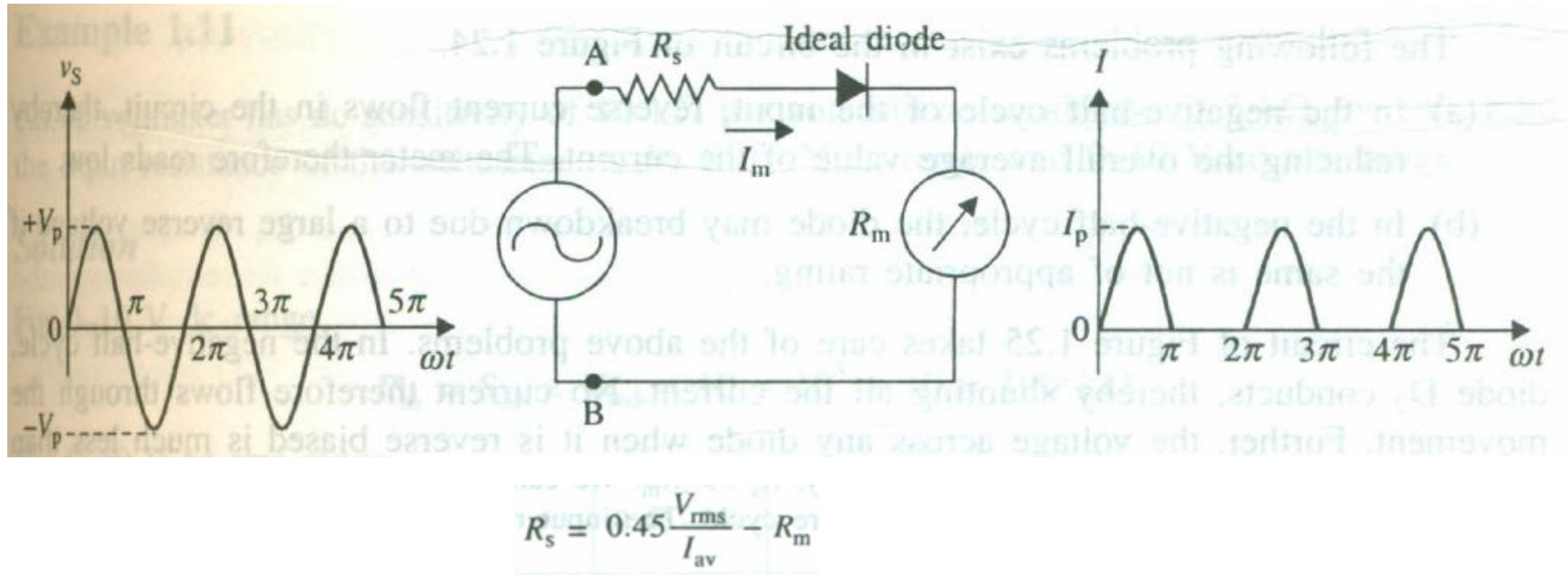
A 50 μA meter movement with an internal resistance of 50 Ω is to be used in a 0–50 V rms range in the arrangement shown in Figure 1.24. Find the value of the required multiplier resistance.

Solution

$$R_s = 0.45 \frac{V_{\text{rms}}}{I_{\text{av}}} - R_m = 0.45 \frac{50}{50 \times 10^{-6}} - 50 \cong 450 \text{ k}\Omega$$

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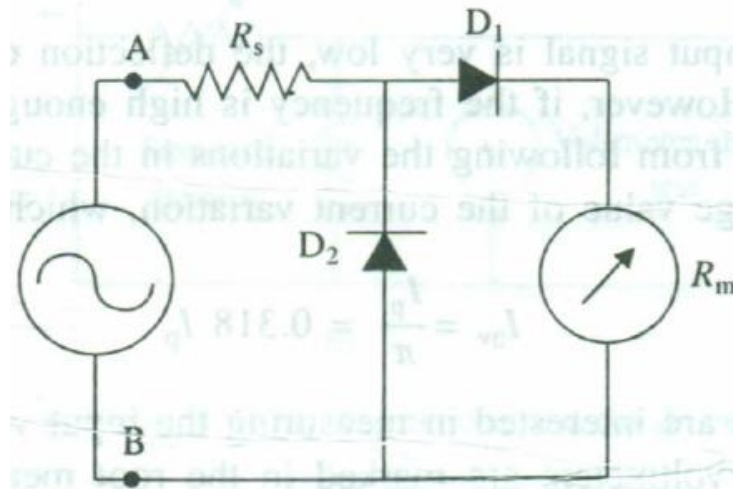
Half-wave Rectifier AC Meters



Disadvantages:

1. In negative half-cycle, reverse current flows through the circuit
→ reduces average value of current → meter reads lower than actual.
2. High peak inverse voltage (PIV) rating of Diode is required

Solution to the previous problems:



1. In negative half-cycle, D_2 conducts $\rightarrow$ no negative current flows through the meter movement.
2. PIV rating requirement is much less than the large peak input voltage (V_p).

Sensitivity of the voltmeter

Input resistance of this circuit

- $R_s + R_m$ (during the positive half-cycle)
- R_s (during the negative half-cycle)

Generally, $R_s \gg R_m$

And, we know that:

$$R_s = 0.45 \frac{V_{rms}}{I_{av}} - R_m$$

So,

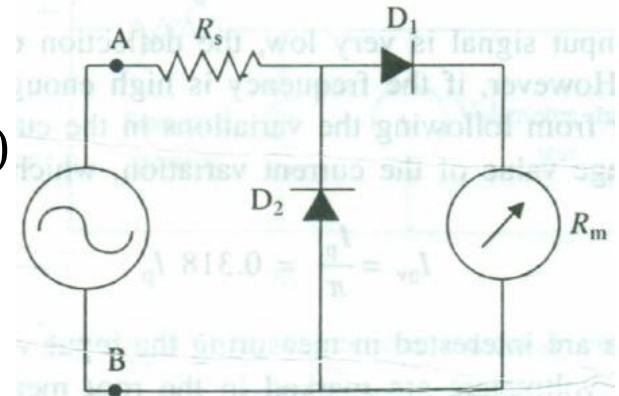
$$R_{in} = 0.45 \frac{V_{fsd}}{I_{fsd}} = S_{ac} V_{fsd}$$

I_{fsd} is the full-scale deflection current of the movement

V_{fsd} is the full-scale range (rms) voltage.

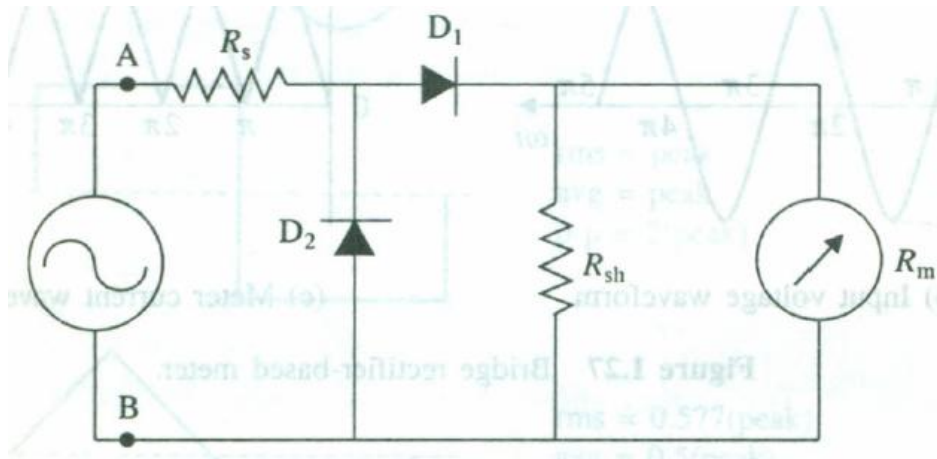
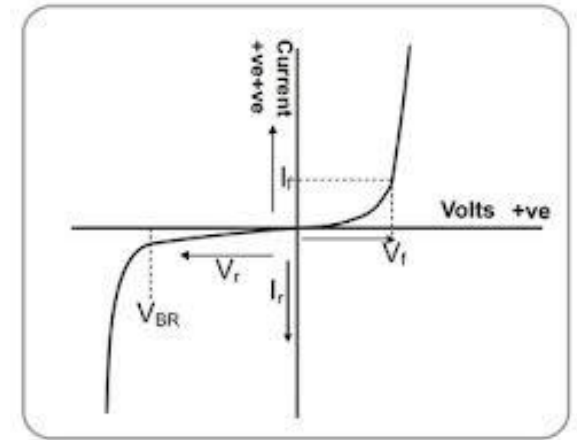
S_{ac} is the AC sensitivity of the voltmeter and is defined as:

$$S_{ac} = \frac{0.45}{I_{fsd}}$$



Non-linearity in the lower rms ranges

- Diode I - V characteristics is non-linear
- This non-linearity is maximum at low values of current
- Effect of non-linearity can be reduced by operating the diode at relatively high current

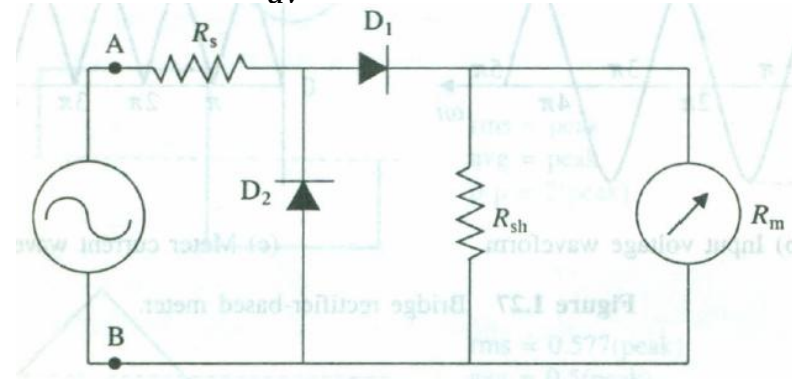


Improving the linearity

If $R_{sh} = R_m$, the current through diode D_1 doubles to $2I_{av}$.

So, we can write:

$$R_s = 0.45 \frac{V_{rms}}{2I_{av}}$$

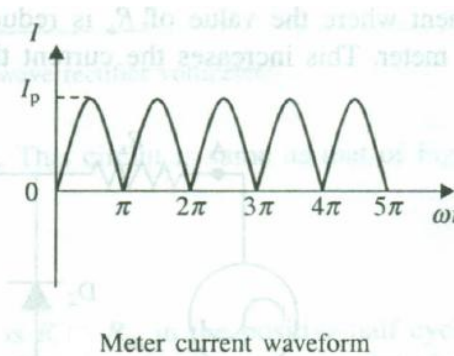
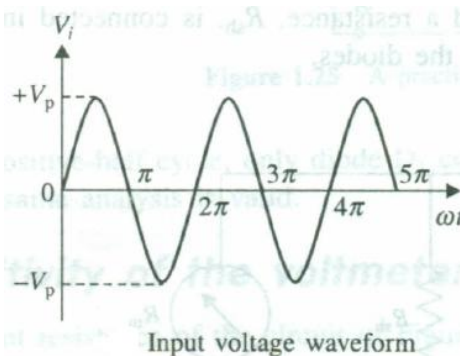
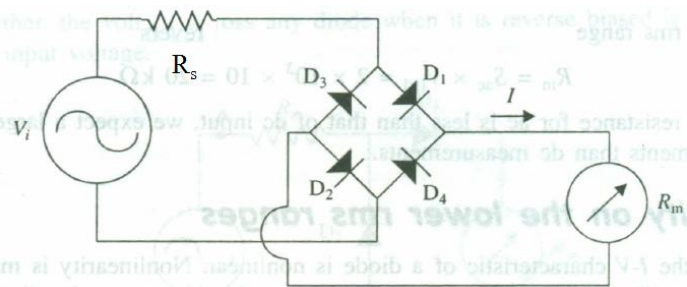


The AC sensitivity is now defined as:

$$S_{ac} = \frac{0.45}{2I_{fsd}}$$

So, S_{ac} is further reduced.

Full-wave Rectifier-based Meter



The average current is given by, $I_{av} = \frac{2I_p}{\pi} = 0.636I_p$

R_s is given by, $R_s = 0.9 \frac{V_{rms}}{I_{av}} - R_m$

Now, the AC sensitivity becomes: $S_{ac} = \frac{0.9}{I_{fsd}}$

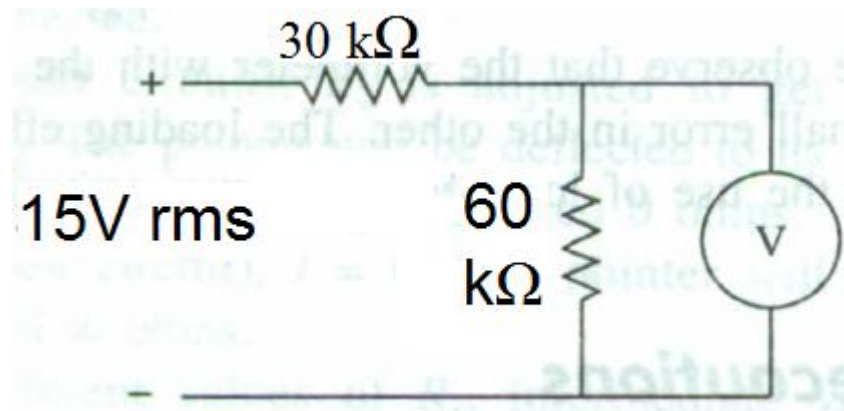
This is double that of a half-wave rectifier-based meter.

Problem Exercise 1

The ac voltmeter is used to measure voltage V across $60\text{ k}\Omega$ resistor in the following circuit. What is the minimum voltage reading that should be observed?

The parameters of the ac voltmeter are:

- Full-wave rectification
- $100\text{ }\mu\text{A}$ meter movement
- 15V range
- Accuracy = $\pm 3\%$ fsd

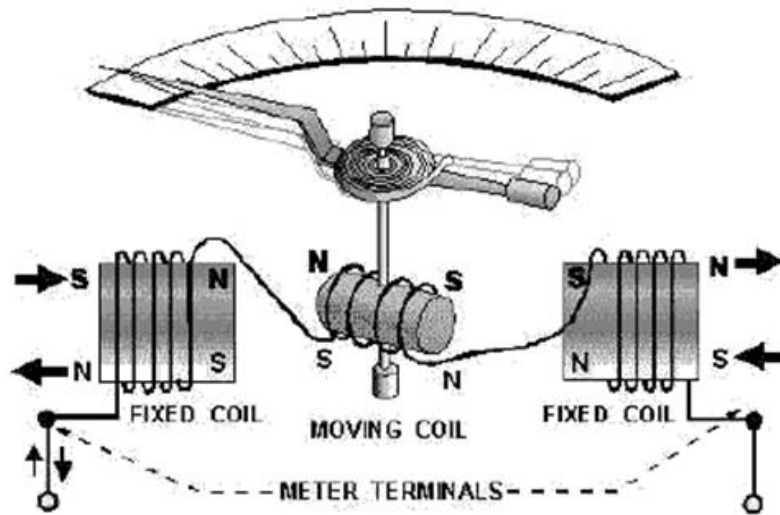


L-6

MEASUREMENT OF POWER AND ENERGY

Electro-dynamic/Dynamometer Wattmeter Principle

RECAP: The basic electrodynamic meter construction for use in voltmeter and ammeter

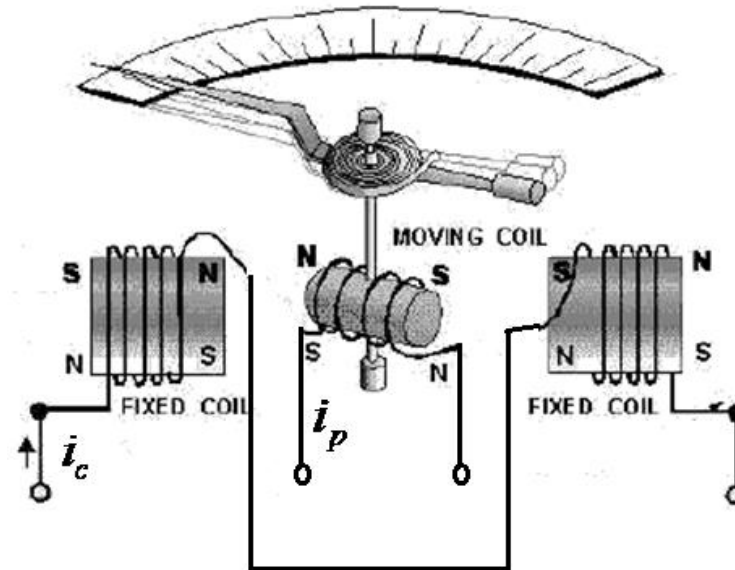


The fixed coils and the movable coil connected in series; same current flows through them

We had before: $T_d \propto i^2$, or, $\theta \propto i^2$

For rapidly varying i , we know that $\theta_f \propto (i^2)_{av}$, Or, $\theta_f \propto \frac{1}{T} \int_0^T i^2 dt$

When used as a WATTMETER, the coils are connected in a different way, as voltage and current, both, are responsible for power.



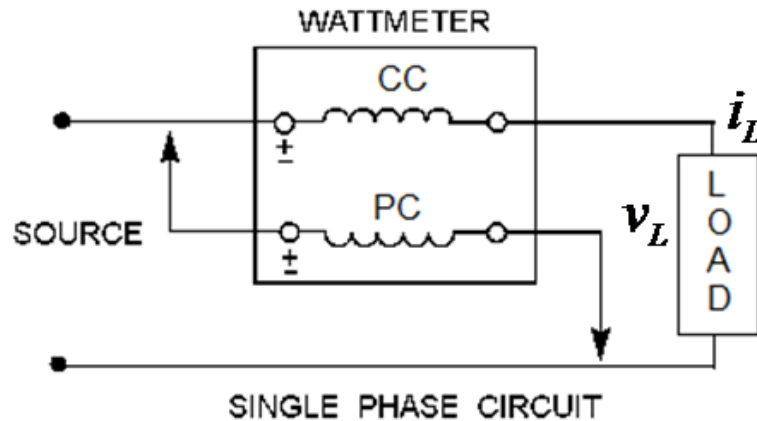
Fixed Coils: Connected in series, effectively work as a single coil (**Current Coil**)

Movable Coil: Connected in series with a high-resistance resistor, together work as **Potential Coil** (also called **Pressure Coil**)

Now,

$$\theta_f = K \frac{1}{T} \int_0^T i_c i_p dt$$

Wattmeter Connection for Metering:



1. The CC is connected in series with the load/line to carry the total load/line current

$$\text{So, } i_c = i_L$$

2. The PC is connected across the load to sense the voltage in its proportion and **in phase** with it

$$\text{So, } i_p = \frac{v_L}{R_p}$$

R_p : Total resistance of the PC (self inductance, L, is negligible)

The deflection equation now becomes:

$$\theta_f = K_1 \frac{1}{T} \int_0^T v_L i_L dt$$

By definition, the average power in a circuit/consumed by a load is

$$P_{av} = \frac{1}{T} \int_0^T v_L i_L dt$$

That means, $\theta_f \propto P_{av}$

So, the electrodynamicometer deflection linearly varies with the average power of the load/circuit.

The above Power equation is valid **irrespective of waveforms**. So
Electrodynamometer Wattmeter is applicable for **DC, AC (any wave shape)**

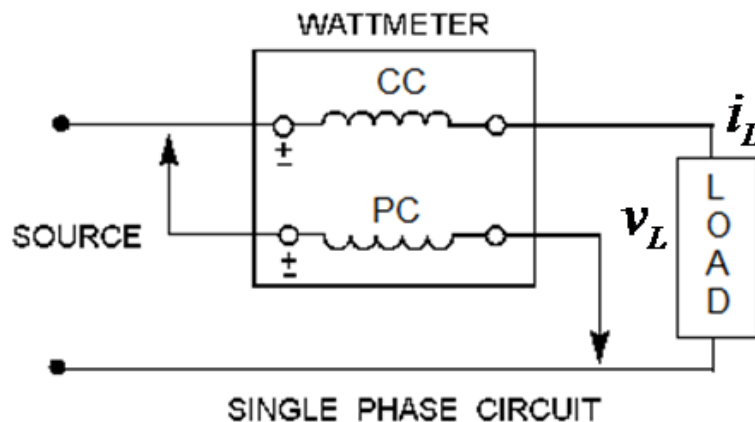
Wattmeter Connections

Wattmeters have one CC terminal and one PC terminal marked, " $\pm$ "

For up-scale reading:

1. The marked terminal of CC should be connected to the incoming line
2. The marked terminal of PC should be connected to the line side in which the CC is connected

If for any reason the meter reads backward (as sometimes happen for two-wattmeter method of measuring 3-phase power), the CC connection should be reversed.



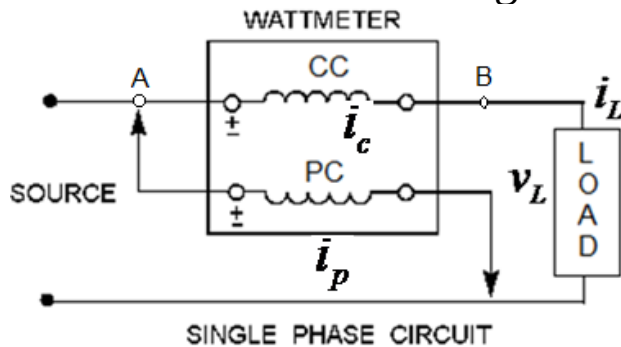
Errors Due to Wattmeter Connections

For correct reading of the load power:

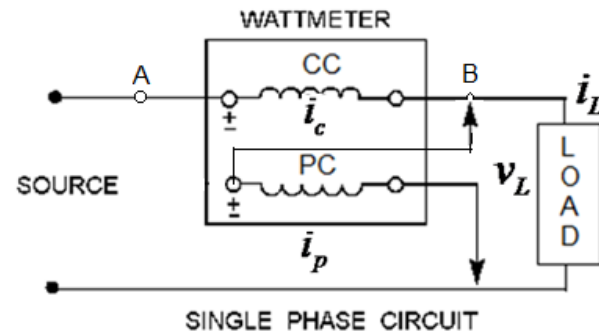
- * the CC should carry exactly the load current; and
- * the PC should be connected across the load terminals

Practically both cannot be achieved together.

- (a) With the PC connected to point 'A', the load current is properly metered, but the voltage across the PC is higher by the amount of the drop across the CC.
- (b) With the PC connected to point 'B', the load voltage is properly metered, but the current through the CC is now greater by the amount i_p .



(a)



(b)

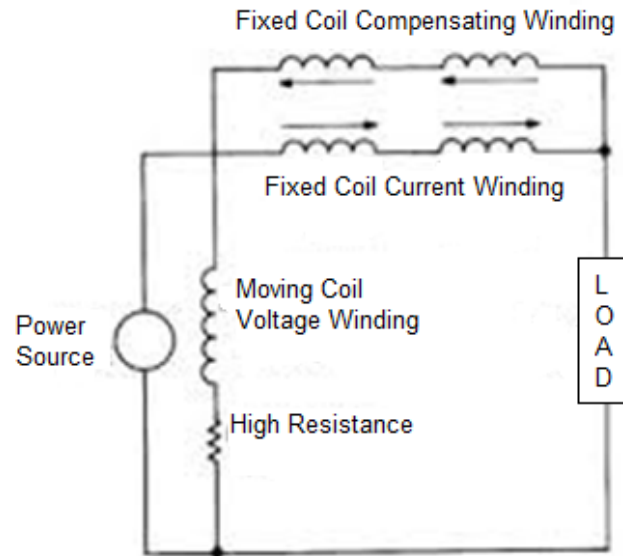
Both cases read higher than the actual power of the load.

Errors Due to Wattmeter Connections

Minimizing connection error:

1. PC connects to point 'A' for low-current, high-voltage loads
2. PC connects to point 'B' for high-current, low-voltage loads

Compensated Wattmeter:



- Two-oppositely wound fixed coils are employed.
- One winding using heavy wire carries the total current of $(i_L + i_p)$ (Connection 'B')
- Second winding using thin wire carries only the potential coil current, i_p
 - Fluxes produced by these two windings are opposite to each other. So, effect of i_p is cancelled.

Errors Due to Wattmeter PC Self-Inductance

Why a **High-Resistance** resistor is connected in series with the Potential Coil (PC)?

- Primarily, to make the PC circuit as more resistive as possible, so that the current in the PC is practically **in phase** with the voltage.
- Practically, the inductance of PC may not be negligible and a correction factor may be required for true power

Instrument Transformers

Introduction

- Called ‘Instrument Transformer’ as they are mainly used for instrumentation of the measurement of high-power system parameters, e.g. voltage, current, power, energy, power factor and frequency.
- Used to **step down** the voltage/current to be measured. Actual measurements are done by other measuring instrument, like voltmeter, ammeter, wattmeter, energymeter.
- Have wide applications in **protection circuits** of power systems, e.g. over-current, under-voltage, earth fault and various other kinds of relays

Types:

1. **Current transformer (CT):** Used to transform high current of a power line to a lower value suitable for direct connection to an AC ammeter
2. **Potential Transformer (PT):** used to transform high voltage of a power line to a lower value suitable for direct connection to an AC voltmeter

Functions of an Instrumentation Transformer

Performs two important functions:

1. Extend the range of AC measuring instrument, similar to the shunt or the multiplier resistors that extends the range of a DC meter.
 - CT and PT provide required range extensions through their transformation ratio.
2. Serve to isolate the measuring instrument from high-voltage power line
 - It is impractical to bring the several-hundred kV high-voltage lines directly to an instrument panel because of the safety hazards involved and the insulation problems of high-voltage lines running closely together in the confined space of the instrument panel.
 - CT, PT allows low-voltage, low-current wires from the transformer secondary brought to the instrument panel, thereby minimizing safety hazards and insulation problems.

Ratios of Instrument Transformer

Actual Transformation Ratio (R):

$$R = \frac{\text{Magnitude of actual primary current}}{\text{Magnitude of actual secondary current}} \quad \dots \text{ For C.T.}$$

$$R = \frac{\text{Magnitude of actual primary voltage}}{\text{Magnitude of actual secondary voltage}} \quad \dots \text{ For P.T.}$$

Nominal Ratio (K_n):

$$K_n = \frac{\text{Rated primary current}}{\text{Rated secondary current}} \quad \dots \text{ For C.T.}$$

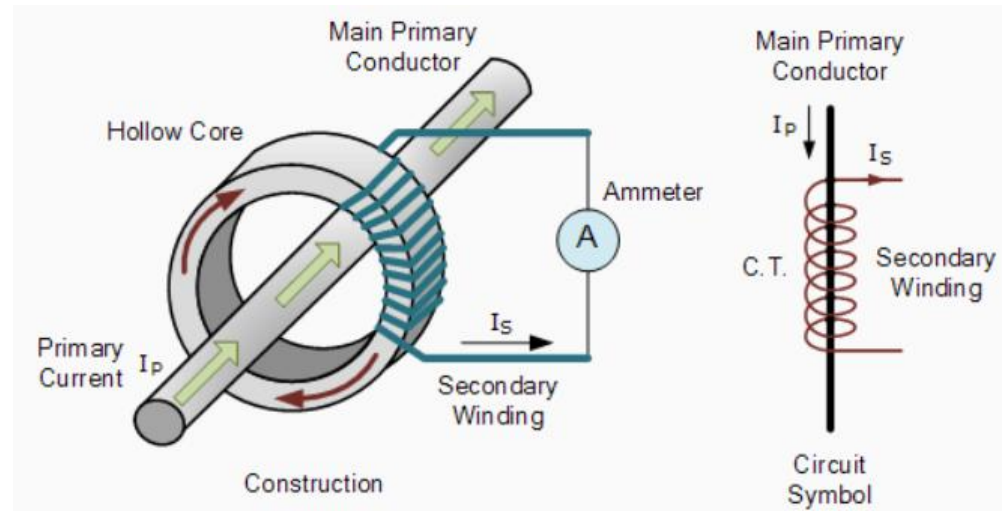
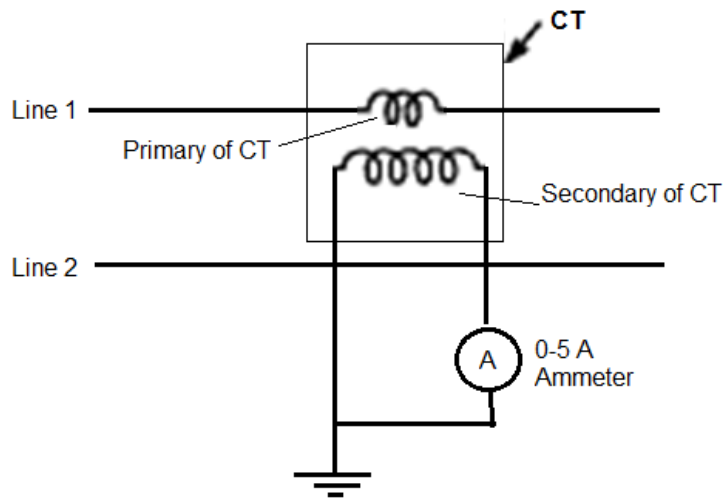
$$K_n = \frac{\text{Rated primary voltage}}{\text{Rated secondary voltage}} \quad \dots \text{ For P.T.}$$

Turn Ratio (n):

$$n = \frac{\text{Number of turns of secondary winding}}{\text{Number of turns of primary winding}} \quad \dots \text{ For C.T.}$$

$$n = \frac{\text{Number of turns of primary winding}}{\text{Number of turns of secondary winding}} \quad \dots \text{ For P.T.}$$

CT



Construction and special design consideration for CT:

1. Its construction is very different from the normal power transformer
2. In most cases, the primary is only one turn or a single conductor connected in series with the load/line whose current is to be measured. The secondary winding has a larger number of turns and is connected in series with the meter.
3. Often the primary winding is a single conductor in the form of a heavy bar of copper or brass running through the core of the transformer. Such a CT is called a bar-type CT.

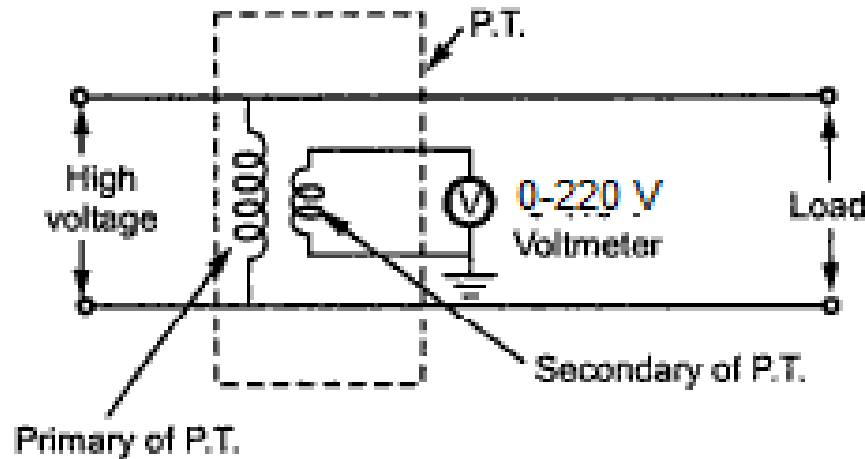
4. The secondary is generally designed to deliver a current of 5 A. An 800/5-A bar-type CT would have 160 turns in the secondary coil.
5. Power rating of the windings is low, as they are not meant to supply any load other than the instruments losses
6. Very low voltages are involved with primary and secondary windings. So unlike PT, high insulation is not required for CT. So its construction is very simple

Why the secondary of a CT should not be open while the primary is energized?

When the secondary is open-circuited, the voltage developed across the open secondary terminals may be very high, because of the step-up ratio. Since a CT is not designed for high insulation to sustain a high voltage like this, it can easily break down the insulation of the CT

The secondary terminal of a CT should therefore always be short-circuited, or connected to a meter or relay coil

PT



Construction and special design consideration for PT:

1. Construction is similar to the normal power transformers
2. Extremely accurate ratio step-down transformer. Secondary is usually rated for 230 V irrespective of the primary voltage rating.
3. Low power winding rating, as they are not meant to supply any load other than the instrument losses
4. For very high primary voltages, the insulation between the primary and secondary windings must be able to withstand large potential difference.

L-8

TRANSDUCERS

Introduction

- Process of converting one form of energy into another is called ‘Transduction’, and the device that performs this function is called a ‘Transducer’.
- Physical parameters to be measured are most of the time ‘non-electrical’.
- Non-electrical quantities are converted into electrical quantities for better measurements. This is because electrical methods of measurement (sensing, signal conditioning, transmission, indication) are much more advantageous than other methods (e.g. mechanical and chemical).

- The devices which convert other form of energy into electrical forms are called ‘Electrical Transducers’
- However, with reference to this course, when we say ‘transducer’, it refers to ‘electrical transducer’.

A nice definition:

‘A transducer is a device which, when actuated by energy in one form, produces an electrical output energy ’

Advantages of Electrical Transducers

The advantages of converting physical quantities into analogous electrical quantities are:

1. Electrical amplification and attenuation can be done easily and with static devices
2. The mass-inertia effects are minimized.
3. Effects of friction are minimized
4. Can be controlled with a very small power level
5. Output can be easily used, transmitted, stored and processed
6. Telemetry is used in almost all sophisticated measurement systems. The entire aerospace research and development is based upon telemetry and remote control. Mechanical transmission would not make this possible.
7. Miniaturization is possible by the use of ICs

Classification of Transducers

Can be classified according to their applications, method of energy conversion, nature of output signal and so on.

- ✓ A classification can be done based on the **electrical circuit parameter** that is meant to be **changed** due to the presence of the physical parameter to be measured:

1. Resistance
2. Capacitance
3. Inductance
4. Voltage and Current

Another Classification of Transducers

Primary transducers

Transducers that sense a physical phenomenon at the first stage of detection; e.g. thermocouple

Secondary transducers

The physical phenomenon is first sensed by a non-electrical transducer. It's output is then converted into an electrical signal by an electrical transducer. This second transducer is called secondary transducer.

For example: in one type of pressure meter, a mechanical tube (Bourdon tube) first converts the pressure to be measured into an analogous displacement. It is then sensed by an electrical transducer which is called secondary transducer

Resistive Transducers

Transducer that employs change in resistance of a sensing element is called resistive transducer.

Resistive transducers are preferred over transducers employing other principles.

The well-known equation of resistance of a resistor element: $R = \frac{\rho L}{A}$

Design basis: **vary any combination of the 3 quantities**

For example: temperature can change ρ ; strain can change ρ , L and A , *altogether*

Applications:

Temperature, strain, displacement, force, pressure etc.

Strain Gauge

If a metal conductor is **stretched** or **compressed**, both L and A change, so its R changes.

Also ρ changes when it is strained. This property is called **piezoresistive** effect.

Resistance strain gauges are also known as **piezoresistive gauges**.

Theory of Strain Gauges:

Resistance of an unstrained wire,
$$R = \frac{\rho L}{A}$$

Let a tensile stress s be applied to the wire. This causes a positive longitudinal strain and negative lateral strain.

Let, ΔL = change in length, ΔA = change in area, ΔD = change in diameter and ΔR = change in resistance

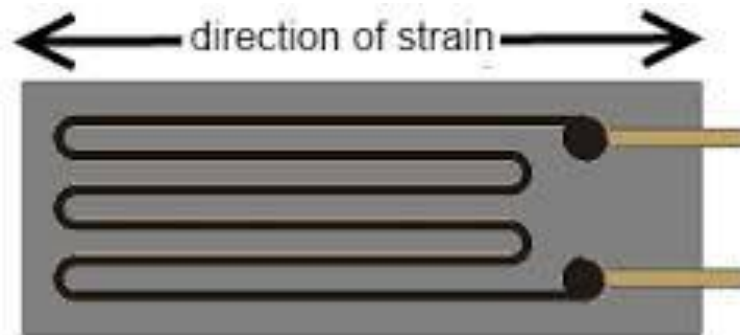
If the piezoresistive effect is negligible, the gauge factor is, $G_f = 1 + 2\nu$

The Poisson's ratio for all metals is between 0 and 0.5

Unit of strain is **microstrain**

1 microstrain = $1\mu\text{m/m}$

Many types of strain gauges are there. A **bonded wire strain gauge** is shown here.



Measurement of resistance: Done by bridges (e.g. Wheatstone bridge)

Applications

For example:

1. Structural stress analysis

- e.g., smart bridges. In order to keep bridges in good conditions, people use strain gauge sensors on bridges to monitor stresses on weak points of a bridge.
- A bridge can have hundreds of sensors to supervise its stress levels at hundreds of points of the bridge



2. Measurement of force, pressure, flow

- indirectly measures the force, pressure through the material strains. Mechanics of materials provide the necessary equation between strains and stresses.
- First measure differential pressure in a section of pipe/tube/channel. Then Hydrodynamics gives the necessary equation between differential pressure and flow rate

L-9

Resistance Temperature Detector (RTD) Or Resistance Thermometer

Resistance-Temperature Characteristics

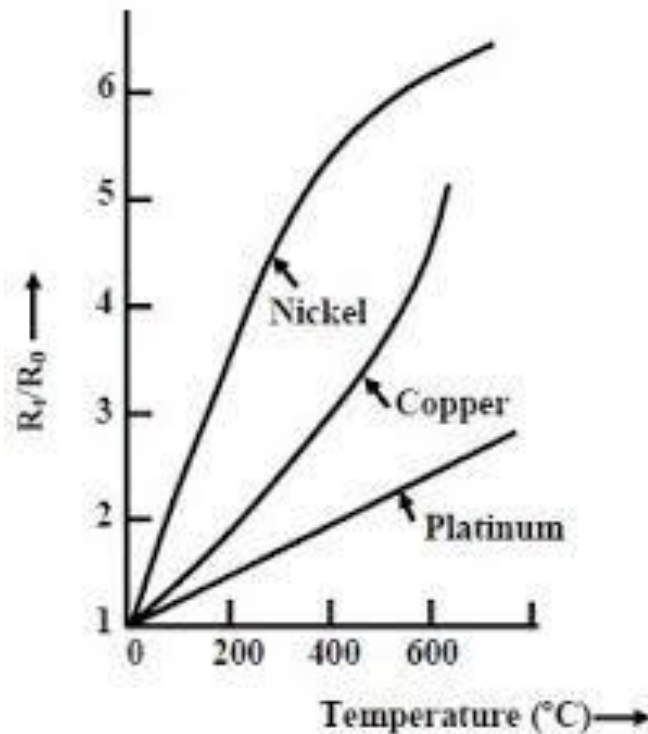
- ❖ RTD uses the change in resistance of **a METAL CONDUCTOR** to determine the temperature.
- ❖ The **resistivity** of metal conductor changes with temperature, and so does the resistance
- ❖ Variation of resistance ***R*** with temperature ***T*** (°K) can be represented by the following well-known relationship:

$$R = R_0(1 + \alpha_1 T + \alpha_2 T^2 + \dots + \alpha_n T^n + \dots)$$

Where, R_0 = resistance at temperature $T = 0$

and, $\alpha_1, \alpha_2, \alpha_3$ are constants

- ❖ Resistance-Temperature characteristics of **commonly used** RTD metals are plotted (not-exact) here:



- The curve shows that the curves are nearly linear, or at least can be approximated as linear for short temperature spans.

Table: Metals used for RTD

Metal	Resistance Temperature Co-efficient (/°C)	Temperature range (°C)	Melting point (°C)
Platinum	0.39	(-260) – (110)	1773
Copper	0.39	(0) – (180)	1083
Nickel	0.62	(-220) – (300)	1435
Tungsten	0.45	(-200) – (1000)	3370

❖ Desirable characteristics of an RTD material:

- Change in resistance per unit change in temperature should be as large as possible
- High resistivity, so that a minimum volume of material is used for the construction of RTD
- The R versus T curve should be continuous, stable and repeatable.

❖ **Platinum (Pt)** is specially suited for use in RTD (Known as **PRTD**).

- Withstand high temperatures while maintaining excellent stability
- As a noble metal, it shows limited susceptibility to contamination

- ❖ **Gold, Silver:** rarely used, as they have low resistivity
- ❖ **Tungsten:** relatively high resistivity, but reserved for high temperature applications as it is extremely brittle and difficult to work with.
- ❖ **Cu:** low resistivity forces the element to be longer than the Pt element; linearity and low cost make it an economical alternative.
- ❖ The most common RTDs are made of either **Pt, nickel or nickel alloys.**
- ❖ For measurement integrity (**precision and accuracy**), **Pt** is the best choice

Equation: Linear Approximation

- ❖ For some metals, linear relationships can exist over a specified span of the curve

$$R_{\theta} = R_{\theta_0} (1 + \alpha_{\theta_0} \Delta\theta) \quad \text{for } \theta_1 \leq \theta \leq \theta_2$$

Where, R_{θ} = Approximate resistance at temperature θ °C;

R_{θ_0} = Approximate resistance at temperature θ_0 °C;

$\Delta\theta = \theta - \theta_0$ = Change in temperature, °C

α_{θ_0} = Resistance temperature co-efficient at temperature θ_0 °C

- ❖ Two unknowns are here, so two points in the line are necessary to solve them.
- ❖ The equation is only applicable for the **specified span** of the temperature.

Equation: Quadratic Approximation

- ❖ A quadratic approximation of the Resistance-Temperature curve can be more accurate over a limited range of temperature. It has both **1st** and **2nd** order temperature terms.

$$R_{\theta} = R_{\theta_0} \left[1 + \alpha_1 \Delta\theta + \alpha_2 (\Delta\theta)^2 \right] \quad \text{for } \theta_1 \leq \theta \leq \theta_2$$

Where, α_1 = Linear fractional change in resistance, /°C

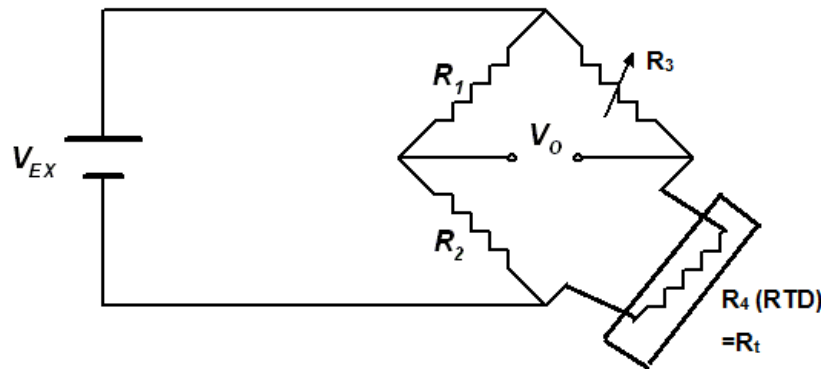
α_2 = quadratic fractional change in resistance, /(°C)²

- ❖ Values of the three unknowns (α_1 , α_2 and R_{θ_0}) can be found from tables or graphs by using three points.

Measurement of RTD Resistance

Measurement Bridges are used for this purpose.

Wheatstone Bridge:



- RTD is connected as one of the arms of the bridge.
- So,
$$R_t = R_3 \left(\frac{R_2}{R_1} \right)$$
- Potentiometer R_3 can be calibrated in terms of temperature directly.

Applications

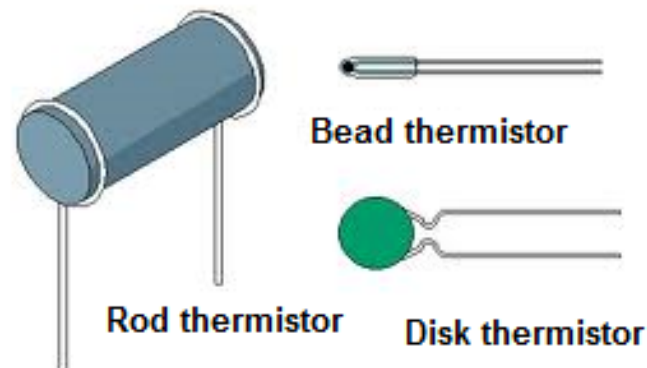
- ❖ RTD is one of the most **ACCURATE and PRECISE temperature sensors**.
- ❖ Most industrial applications require very **high temperatures** for their processes, and they **do not need that much accurate** temperature measurements, e.g. for a 250 °C temperature process, it may not matter whether it is 248 °C or 252 °C.
- ❖ **RTD Applications:**
 - **Bio-reactor:** here biological cells are grown in a controlled manner. A deviation of temperature from the set value can cause the cell to die or grow defectively. So, precise and accurate temperature measurement is highly required for exact control of bioreactor temperature.
 - **Low temperature process:** where small change of temperature needs to be measured with repeated accuracy

Thermistors

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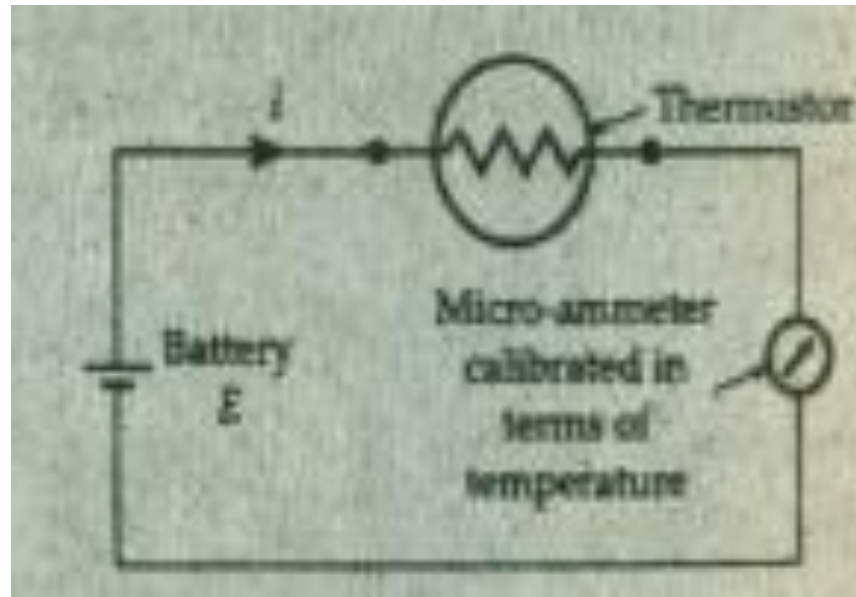
General

- ❖ Contraction of the term, “thermal resistors”
- ❖ Generally made of **semiconductor** materials
- ❖ Have negative temperature co-efficients, i.e. resistance decreases with increase in temp (Due to production of **higher number of electron-hole pair** generation as temperature increases)
- ❖ **sensitivity is high**, so suitable for very small changes in temperature.
 - This makes them useful for **precision temperature measurement** applications.
- ❖ Relatively low cost and small size.
- ❖ Measurement range: approximately, **(-60) – (15) °C**



❖ Measurement and control of temperature:

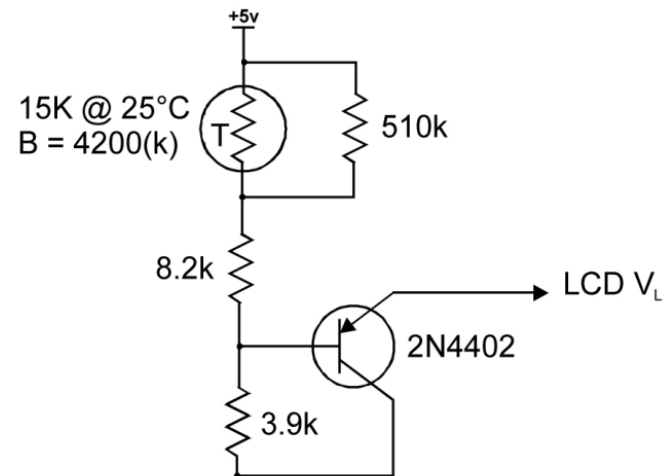
- Provides a large change in resistance for a small change in temperature (**high sensitivity**)
- Can be connected in a simple series circuit of a battery and a micro-ammeter. The micro-ammeter can directly be calibrated in terms of temperature.
- A thermistor can be connected as one arm of a bridge circuit for more sensitive measurement.



Applications

❖ Temperature compensation:

- ❖ Thermistors have negative temperature co-efficients, opposite to that of electrical resistors. So, they are widely used to compensate for the effects of temperature.
- ❖ They are so connected in the circuit that the total circuit resistance is constant over a wide range of temperatures



❖ Other Applications:

- ❖ Measurement of flow, providing time delay etc.